Homework 6 - Part 1a: Incompressible Taper-Flat Slider

Results

For Part 1a, the pressure distribution for the 2D incompressible taper-flat slider bearing was computed using the Finite Element Method. The load F and the center-of-pressure x_p were calculated using the following integrals over the domain Ω:

$$F = \int_\Omega p(x, y)\, dx\, dy$$
$$x_p = \frac{1}{F} \int_\Omega x\, p(x, y)\, dx\, dy$$

Using the numerical solution, the computed values are:

- **Computed Load (F):** 0.2066599 N
- **Center of Pressure (x_p):** 0.002208667 m

Analytical Verification

The analytical load $\tilde{F}$ for a 1D pure wedge configuration is given by:

$$\tilde{F} = \frac{\Lambda L W}{(1-\alpha)^2} \left(\ln(\alpha) + 2\frac{1-\alpha}{1+\alpha} \right)$$

where $\alpha = h_1/h_2$ and $\Lambda = 6\mu U L/h_2^2$.

Substituting the given parameters, we obtain:

- **Analytical Load ($\tilde{F}$):** 8.154767 N

Note: The significant difference between the numerical load (0.207 N) and the analytical load (8.15 N) is expected. The numerical model simulates a 2D slider with a 1° leading taper, whereas the analytical formula assumes a 1D, perfectly flat, full-length wedge.

Pressure Distribution

The numerical pressure distribution $p(x, y)$ over the slider domain is shown in Figure 1.

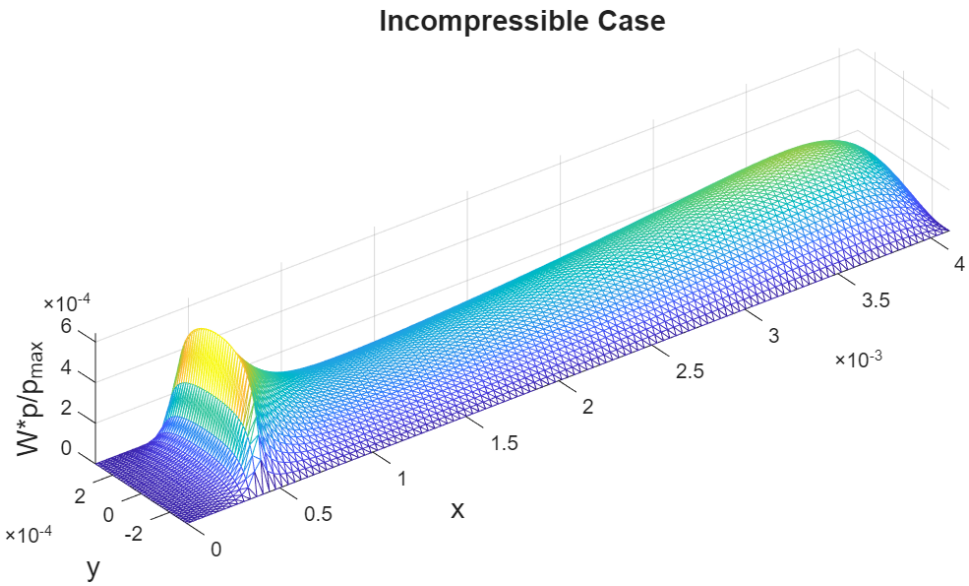

Figure 1: Normalized pressure distribution for the incompressible taper-flat slider bearing.